

BSc Electronics & Computer Maintenance

CNC PLOTTER MACHINE

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The Process of this Project

This CNC machine is powered by using two stepper motors and one servo motor. That is, it acts across the X, Y, and Z axes. Also, all of these motors are activated using the L293D motor drive shield and Arduino UNO board. If we want to draw a character through this CNC machine, its G-code must be obtained. The Inkscape software has been used for this. Also, the Arduino program sets the CNC machine to work, and a processing program is used to run the G-code. In this tutorial, how to do all of these are described step by step and by the end of the tutorial, you can easily create a CNC machine at home.

Components Required

Arduino UNO board
L293D motor driver shield
CD ROM/DVD writer
Servo motor
Wires
Nut and bolt
Pen
Spring
Foam board/Cardboard

Circuit Diagram



Block Diagram



CNC Machine Merits

1.The CNC machine will be able to execute its operations with high accuracy and precision for a long period of time.

2.CNC machines are programmed with a design which can then be manufactured hundreds or even thousands of times. Each manufactured product will be exactly the same.

3.Less skilled/trained people can operate CNC machines unlike manual lathes / milling machines etc. which need skilled engineers.

4.CNC machines can be updated by improving the software used to drive the machines

5.Training for correct use of CNC machines is available through the use of 'virtual software'. This software is like a computer game that allows the operator to practice using the CNC machine on the screen of a computer.

Conclusion

In this paper we have presented the concept of a low cost two-axis mini cnc CNC plotter. The existing CNC machines are of high cost, difficult to maintain and requires highly skilled operators. Our CNC plotter overcomes these problems. It is of low cost and easy to control and there is no need of highly skilled operators. It can be used for long hours at a stretch which is not possible in existing ones. It is hoped to extend this work for future development.